

Input f (3×3)

f_{11}	f_{12}	f_{13}
f_{21}	f_{22}	f_{23}
f_{31}	f_{32}	f_{33}

Kernel g (2×2)

g_{11}	g_{12}
g_{21}	g_{22}

1. Cross-Correlation ($f \star g$)

Kernel slides in its **original orientation**. No padding.

f_{11} g_{11}	f_{12} g_{12}	f_{13}
f_{21} g_{21}	f_{22} g_{22}	f_{23}
f_{31}	f_{32}	f_{33}

Output

$f_{11} \cdot g_{11} + f_{12} \cdot g_{12} + f_{21} \cdot g_{21} + f_{22} \cdot g_{22}$	$f_{12} \cdot g_{11} + f_{13} \cdot g_{12} + f_{22} \cdot g_{21} + f_{23} \cdot g_{22}$
$f_{21} \cdot g_{11} + f_{22} \cdot g_{12} + f_{31} \cdot g_{21} + f_{32} \cdot g_{22}$	$f_{22} \cdot g_{11} + f_{23} \cdot g_{12} + f_{32} \cdot g_{21} + f_{33} \cdot g_{22}$

2. Full Convolution ($f *_{\text{full}} g$)

Full zero-padding. Kernel is **flipped 180°** and slides starting from edges.

g_{22}	g_{21}	0	0	0
g_{12}	g_{11}	f_{12}	f_{13}	0
0	f_{21}	f_{22}	f_{23}	0
0	f_{31}	f_{32}	f_{33}	0
0	0	0	0	0

Output

$f_{11} \cdot g_{11}$	$f_{11} \cdot g_{12} + f_{12} \cdot g_{11}$	$f_{12} \cdot g_{12} + f_{13} \cdot g_{11}$	$f_{13} \cdot g_{12}$
$f_{11} \cdot g_{21} + f_{21} \cdot g_{11}$	$f_{11} \cdot g_{22} + f_{12} \cdot g_{21} + f_{21} \cdot g_{12} + f_{22} \cdot g_{11}$	$f_{12} \cdot g_{22} + f_{13} \cdot g_{21} + f_{22} \cdot g_{12} + f_{23} \cdot g_{11}$	$f_{13} \cdot g_{22} + f_{23} \cdot g_{12}$
$f_{21} \cdot g_{21} + f_{31} \cdot g_{11}$	$f_{21} \cdot g_{22} + f_{22} \cdot g_{21} + f_{31} \cdot g_{12} + f_{32} \cdot g_{11}$	$f_{22} \cdot g_{22} + f_{23} \cdot g_{21} + f_{32} \cdot g_{12} + f_{33} \cdot g_{11}$	$f_{23} \cdot g_{22} + f_{33} \cdot g_{12}$
$f_{31} \cdot g_{21}$	$f_{31} \cdot g_{22} + f_{32} \cdot g_{21}$	$f_{32} \cdot g_{22} + f_{33} \cdot g_{21}$	$f_{33} \cdot g_{22}$

For a 4×4 input and 3×3 kernel, full convolution requires padding of $P = K - 1 = 2$. The total padded grid is 8×8 . Output size is $N_{\text{in}} + K - 1 = 6 \times 6$.

g_{33}	g_{32}	g_{31}	0	0	0	0	0
g_{23}	g_{22}	g_{21}	0	0	0	0	0
g_{13}	g_{12}	g_{11}	f_{12}	f_{13}	f_{14}	0	0
0	0	f_{21}	f_{22}	f_{23}	f_{24}	0	0
0	0	f_{31}	f_{32}	f_{33}	f_{34}	0	0
0	0	f_{41}	f_{42}	f_{43}	f_{44}	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0